

POKEMON EVOLVE



In mysterious world of pokemon , there is a pokemon known as CryptoMat and there are two types of stones (water stone and fire stone). CryptoMat is a legendary pokemon, it can evolve to three forms CryptoFame by using fire stone, CryptoWave using water stone and CryptoAir using both stones. Cryp-toMat(s) are located in a matrix of size $m \times m$. Now you are given q queries and in each query you are given type of query, n and type of stone. For the type of stone 0 for fire stone and 1 for water stone. For each type of query you will be given 0 and 1 where 0 means give a stone to all the CryptoMat in n th row and 1 means give a stone to all CryptoMat in n th column.

Print the total number CryptoMat, CryptoFame, CryptoWave and CryptoAir as given in Input Output.

Other Things to keep in mind

- 1) Fire + Fire = Fire
- 2) Water + Water = Water
- 3) Water + Fire + Anything = Fire + Water (CryptoAir)

Input Format

The first line of input contains m and q where size of the given matrix is $m \times m$ and q is the number of queries. Then next q lines contain 3 integer where 1st integer represents row(0) or column(1) then n where n represent the n th row or column and 3rd integer represents the type of stone 0 for fire stone and 1 for water stone

Constraints

$$1 \leq m \leq 10^5$$

$$1 \leq q \leq 10^6$$

$$0 \leq n \leq m-1$$

Output Format

Print the number of CryptoAir, CryptoFame, CryptoWave and CryptoMat seperated by a space.

Sample Input 0

```
5 4
0 1 0
0 3 1
1 2 0
1 3 1
```

Sample Output 0

```
2 7 7 9
```

Sample Input 1

```
6 5
0 0 1
0 2 0
0 4 0
1 1 0
1 3 1
```

Sample Output 1

```
3 8 13 12
```